

Is culture analysable?

Social and Cultural Analytics (7AAVBCS5)

Dr Mark J. Hill · mark.j.hill@kcl.ac.uk

2026-09-28



Lecture Plan

Part I: What is Social and Cultural Analytics?

- What is culture?
- What is datafied culture?
- Cultural analytics and big data

Part II: Introduction to Python

- Introduction to Python and Positron
- Python as a calculator
- Variables in Python

Goals

First, convince you of the value of the module.

Second, set the substantive scene and encourage questions around social and cultural analytics such as:

- What does it mean to study culture?

- What types of culture can be studied?
- How do you, or can you, study it?

Third, introduce you to Python and Positron with some very basic coding.

What is Culture? Foundational Definitions

What is culture?

‘Culture ... is that complex whole which includes knowledge, belief, art, morals, law, custom, and any other capabilities and habits acquired by man as a member of society.’ (Edward Burnett Tylor, *Primitive Culture* 1870)

‘Culture consists of patterns, explicit and implicit, of and for behaviour acquired and transmitted by symbols, constituting the distinctive achievements of human groups, including their embodiment in artifacts; the essential core of culture consists of traditional (i.e. historically derived and selected) ideas and especially their attached values; culture systems may, on the one hand, be considered as products of action, on the other, as conditional elements of future action.’
(Kroeber, A.L., & Kluckhohn, C. “Culture: A critical review of concepts and definitions.” 1952)

‘Culture is a fuzzy set of basic assumptions and values, orientations to life, beliefs, policies, procedures and behavioural conventions that are shared by a group of people, and that influence (but do not determine) each member’s behaviour and his/her

interpretations of the 'meaning' of other people's behaviour.' (Spencer-Oatey, *Culturally Speaking* 2008)

What is culture? Complex and multifaceted

‘Culture ... is that **complex whole** which includes **knowledge, belief, art, morals, law, custom, and any other capabilities and habits** acquired by man as a member of society.’ (Edward Burnett Tylor, *Primitive Culture* 1870)

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What is culture? Learned and transmitted

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What is culture? Social and shared

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What is culture? Influences behaviour and meaning

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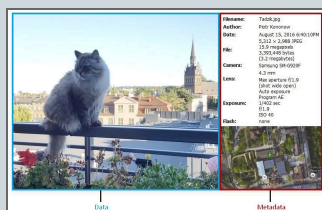
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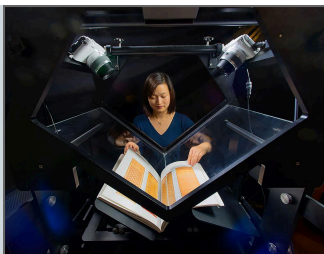
Adding (big) data

How is datified culture different?

How is datified culture different? Abundance

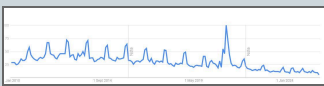
Social interactions, cultural trends, and personal preferences are being quantified and analysed. This quantification allows for a more empirical approach to understanding culture but also raises issues around the interpretation and use of cultural data. (See: Van Dijck 2014; Lupton 2016)



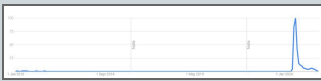


How is datified culture different? Rapid, dynamic and evolving

Digital culture is characterised by its fast-paced evolution aided by the ease of information dissemination. This has led to quicker changes and adaptations in cultural trends and norms. (See: Castells 2010)



GameStop

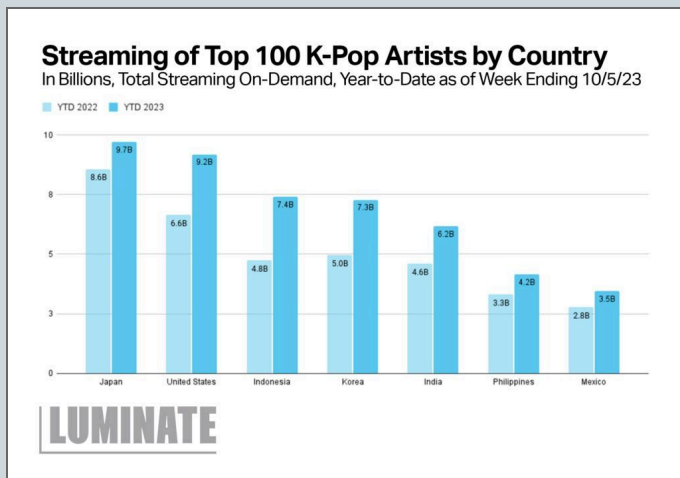


Brat summer



How is datified culture different? Global and interconnective

The internet facilitates global interconnectivity which leads to unprecedented exchange of cultural ideas and practices (See: Appadurai 1996; Graham & Dutton 2019).



“Digital nomads”

Digital culture allows for the formation of communities and identities that are not constrained by geographical boundaries while also allowing people to look at their own (geographic) communities in new ways. (See: Baym 2015)

**WIRED**

Thousands gather in cities across the world to show solidarity with US demonstrations



Apr 6, 2019, 10:51 PM EDT

GABRIEL LEÃO CULTURE DEC 27, 2023 9:08 AM

Perifacon shows the power of gaming, geek culture, and tech outside of uptown São Paulo.

EN





FRANCE

AFRICA

MIDDLE EAST

AMERICAS

HOME

EUROPE

EU far-right parties rally in Madrid behind slogan 'Make Europe Great Again'

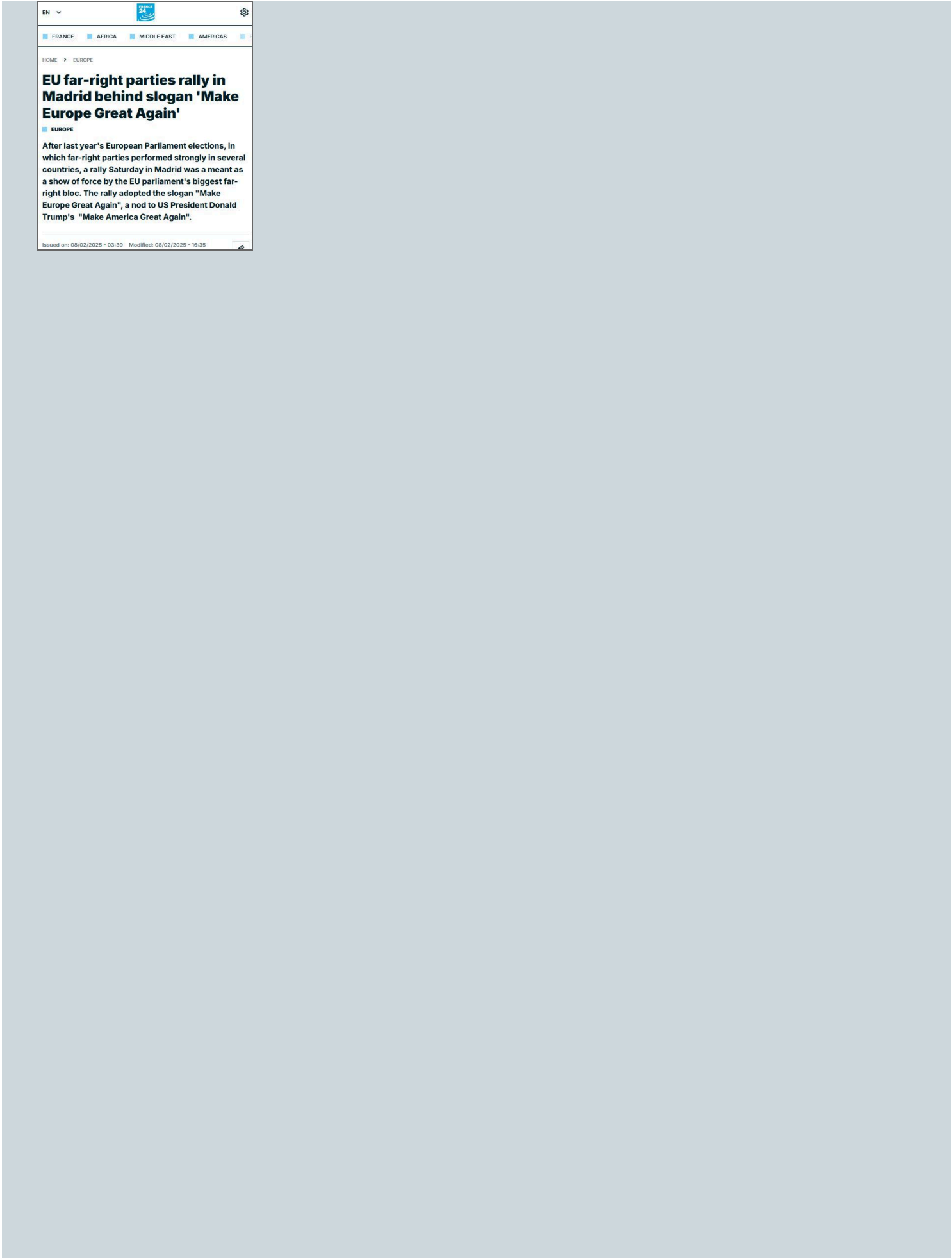
EUROPE

After last year's European Parliament elections, in which far-right parties performed strongly in several countries, a rally Saturday in Madrid was meant as a show of force by the EU parliament's biggest far-right bloc. The rally adopted the slogan "Make Europe Great Again", a nod to US President Donald Trump's "Make America Great Again".

Issued on: 08/02/2025 - 03:39

Modified: 08/02/2025 - 16:35





How is datified culture different? Blurring of public and private spheres

Digital platforms have created spaces where personal experiences are shared widely and publicly, altering what can be considered public and private. (See: Marwick & boyd 2010)



Home assistants

Catching Stray Balls
Football, fandom, and the impact on digital discourse
Mark J. Hill, Kings College London

Research Question & Motivation

Do real-world events trigger toxic behaviour that spreads across unrelated online communities?

Toxicity has been defined as 'interactions directed at an entity designed to be inflammatory' (Daneen et al. 2021).

Why it Matters

- Links toxicating toxic online discourse experience negative psychological consequences (Boukyri et al. 2022; Alvioli et al. 2020)
- Social media amplifies emotional content through platform design (Mili et al. 2025; Kramer et al. 2014)
- Those exposed to emotionally charged content are more likely to express similar sentiments (Beram, and Yang 2013; Brady et al. 2017)

Gap: Limited understanding of how toxicity emerges and moves between communities.

Approach: Football as a 'natural experiment' with clear, timesynced emotional triggers to trace sentiment spillover across digital spaces.

Data & Methods

Scale: 62+ million football posts from 41 football club subreddits (2008-2021)

Matched Events: 2076 match results aligned with posting times and football club subreddit

Cross-Community Analysis: Over half a million pairs of posts by same users in football and one football subreddits (within 15-minute windows)

Match Result	Posts	%
Wise	6,177,963	19.6
Draw	2,693,241	20.6
Losses	3,902,686	20.9

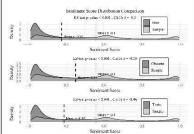
Analysis Pipeline:

1. RedditAPI-based sentiment and toxicity analysis
2. Match outcome correlation with sentiment shifts
3. Cross community sentiment spillover measurement
4. Linguistic feature analysis of negative content

Finding 1: Negative Sentiment & Toxicity

How do we measure toxicity?

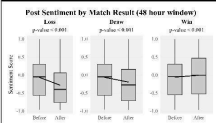
Toxicity is contextual. Posts in football subreddits after a loss may be perceived as negative rather than toxic. That same emotion, however, may be toxic in unrelated football sites. Therefore we use automatic and toxicity detection, as well as proximity lexicons to assess the relationship.



Strong correlation between negative sentiment and problematic content.

Implication: Sentiment is reliable proxy and early warning signal

Finding 2: Football Results Correlate to Online Sentiment

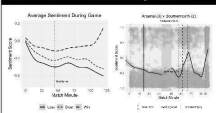


Result	Posts per match	Average ratio	Sentiment
Within 150 Minutes			
Loss	720	0.89**	-0.25**
Draw	784	0.82**	-0.11**
Win	694	1.12**	0.02**
Within 8 Hours			
Loss	577	0.80**	-0.12**
Draw	605	0.84**	-0.09**
Win	581	1.20**	0.00**

Asymmetric Effect: Losses decrease sentiment and posting, win increase posts but lose smaller impact on emotions.

Temporal Correlation: Clear relationship between match events and sentiment shifts

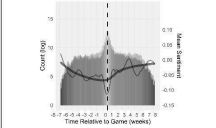
Real-Time Emotional Dynamics



Left: Aggregated sentiment post-match across all matches

Right: Individual match (Arsenal vs. Burnessmouth, 2021)

Below: Post count and sentiment relative to match (using



Key patterns:

- Universal sentiment drop at kick-off (toxicity)
- Diverging patterns based on final match outcome
- Negative reactions more sustained than positive
- Immediate measurable responses to key match events
- Sentiment is more acute to matches
- Sentiment decay over time following matches

Finding 3: Emotional Spillover Across Communities

Emotional states in football subreddits correlate to sentiment in unrelated subreddits

Time Period	Kendall's τ	n
All Match Posts	0.06***	275,863
During Matches	0.115***	234,624
Outside Matches	0.059***	243,839

Finding: Small correlation, but relative changes and statistical significance across large sample represent meaningful patterns of user behaviour at scale. The doubling of τ during matches suggests and would events amplify cross-community emotional spillover. This represents (within this sample) thousands of additional unique posts in unrelated subreddits.

	Negative	Neutral	Positive
Negative	-0.08	-0.76	-0.73
Neutral	-0.02	0.74	-0.59
Positive	-0.07	-0.12	0.66

Finding: Matching sentiment across communities occurs more than chance

Finding 4: Linguistic Spillover of Toxic Features

Method: Measured correlation between linguistic features in paired posts across subreddits.

Feature	Outside Match	During Match	Difference ($\Delta\tau$)
Profanity	0.061***	0.169***	0.108***
Violent words	0.022***	0.049***	0.027***
Interjections	0.059***	0.074***	0.015***
Exclamations	0.114***	0.154***	0.040***
All-together	0.052***	0.139***	0.087***

Finding: All linguistic toxicity markers strengthen during matches

Implication: Heightened emotional states from football events intensify toxic communication patterns across unrelated digital spaces

Implications & Applications

Digital communities are interconnected emotional ecosystems

For Research:

- Methods to measure emotional contagion, spreading
- Application beyond football (elections, breaking news)

For Platform Design:

- Predictive moderation during high risk events
- Early warnings through cross-platforms monitoring
- Indicators for temporally linking cross-community posts

For Society:

- Understanding online-to-online harm pathways
- Hidden mechanisms of toxicity propagation

Broader Impact: Computational evidence of real-world event-driven emotional spillover across unrelated digital communities

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Contact: mark.j.hill@kcl.ac.uk
Pre-print: <https://doi.org/10.48550/arXiv.2206.01612>
Code: <https://github.com/markjhill/2025-catching-stray-balls>

Research Question & Motivation

Do real-world events trigger toxic behaviour that spreads across unrelated online communities?

Toxicity has been defined as 'interactions directed at an entity designed to be inflammatory' (Hanscom et al. 2024).

Why it Matters:

- Users encountering toxic online discourse experience negative psychological consequences (Braghieri et al. 2022; Allcott et al. 2020)
- Social media amplifies emotional content through platform design (Milli et al. 2025; Kramer et al. 2014)
- Those exposed to emotionally charged content are more likely to express similar sentiments (Ferrara and Yang 2015; Brady et al. 2017)

Gap: Limited understanding of how toxicity emerges and moves *between* communities.

Approach: Football as a 'natural experiment' with clear, time-stamped emotional triggers to trace sentiment spillover across digital spaces.

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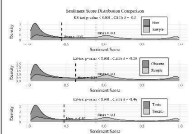
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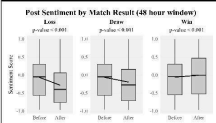
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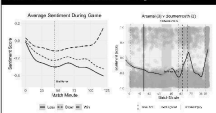


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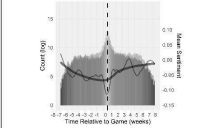
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Finding 4: Linguistic Spillover of Toxic Features

Method: Measured correlation between linguistic features in paired posts across subreddits.

Feature	Outside Match	During Match	Difference ($\Delta\tau$)
Profanity	0.001***	0.009***	0.008***
Violent words	0.002***	0.004***	0.002***
Interjections	0.000***	0.001***	0.001***
Exclamations	0.014***	0.015***	0.002***
All-together	0.002***	0.003***	0.001***

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Matched Events: 20,764 match results aligned with posting times and football club subreddit

Cross-Community Analysis: Over half a million pairs of posts by same users in football and non-football subreddits (within 10-minute windows)

Match Result	Posts	%
Wins	6,477,964	49.6
Draws	2,690,511	20.6
Losses	3,902,686	29.9

Analysis Pipeline:

1. RoBERTa-based sentiment and toxicity analysis
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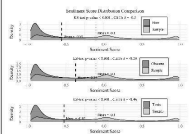
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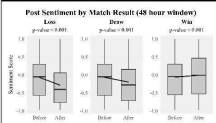
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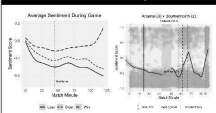


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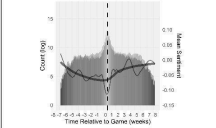
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Neutral	18.92	22.74	-7.59
Positive	-27.49	-5.12	11.66

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- Hidden mechanisms of toxicity propagation

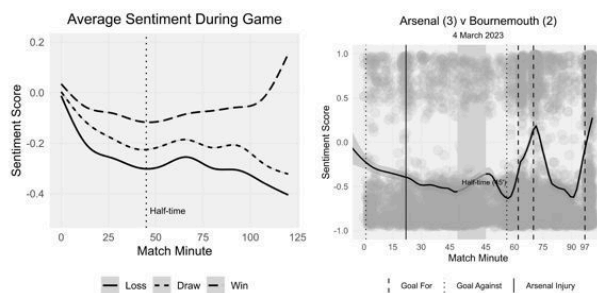
Broader Impact: Computational evidence of real-world event-driven emotional spillover across mediated digital communities

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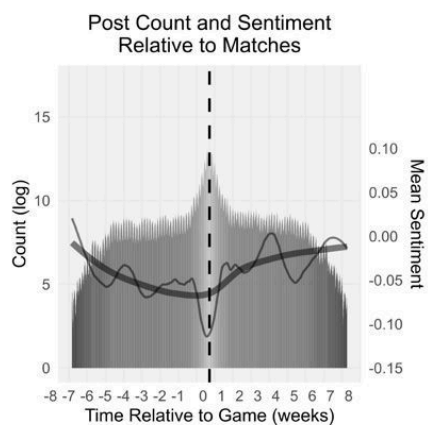
KINGS
LONDON

Contact: mark.j.hill@kcl.ac.uk
Pre-print: <https://doi.org/10.48550/arXiv.2206.01612>
Code: <https://github.com/markjhill/2025-catching-stray-balls>

Real-Time Emotional Dynamics



Left: Aggregated sentiment per-minute across all matches
Right: Individual match (Arsenal vs Bournemouth, 2023)
Below: Post count and sentiment relative to match timing



Key patterns:

- Universal sentiment drop at kick-off (anxiety)
- Diverging patterns based on (final) match outcome
- Negative reactions more sustained than positive
- Immediate measurable responses to key match events
- Sentiment is lower nearer to matches
- Sentiment decay over time following matches

Catching Stray Balls

Football, fandom, and the impact on digital discourse

Mark J. Hill, King's College London

Research Question & Motivation

Do real-world events trigger toxic behaviour that spreads across unrelated online communities?

Toxicity has been defined as 'interactions directed at an entity designed to be inflammatory' (Daneen et al. 2021).

Why it Matters

- Links toxicating toxic online discourse experience negative psychological consequences (Boukyri et al. 2022; Alvioli et al. 2020)
- Social media amplifies emotional content through platform design (Mili et al. 2025; Kramer et al. 2014)
- Those exposed to emotionally charged content are more likely to express similar sentiments (Beram, and Yang 2013; Brady et al. 2017)

Gap: Limited understanding of how toxicity emerges and spreads between communities.

Approach: Football as a 'natural experiment' with clear, timesynced emotional triggers to trace sentiment spillover across digital spaces.

Data & Methods

Scale: 62+ million football posts from 41 football club subreddits (2008-2021)

Matched Events: 20/20 match results aligned with posting times and football club subreddit

Cross-Community Analysis: Over half a million pairs of posts by same users in football and one-football subreddits (within 10-minute windows)

Match Result	Posts	%
Wise	6,177,963	19.6
Draw	2,693,241	20.6
Losses	3,902,686	20.9

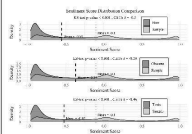
Analysis Pipeline:

1. RedditAPI-based sentiment and toxicity analysis
2. Match outcome correlation with sentiment shifts
3. Cross-community sentiment spillover measurement
4. Linguistic feature analysis of negative content

Finding 1: Negative Sentiment & Toxicity

How do we measure toxicity?

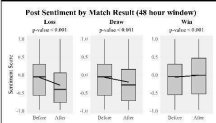
Toxicity is contextual. Posts in football subreddit after a loss may be perceived as negative rather than toxic. That same emotion, however, may be toxic in unrelated football sites. Therefore we use automatic and toxicity detection, as well as proximity lexicons to assess the relationship.



Strong correlation between negative sentiment and problematic content.

Implication: Sentiment is reliable proxy and early warning signal

Finding 2: Football Results Correlate to Online Sentiment

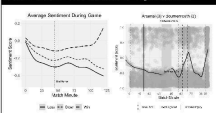


Result	Posts per match	Average ratio	Sentiment
Within 100 Minutes			
Loss	720	0.89**	-0.25**
Draw	714	0.82**	-0.11**
Win	691	1.12**	0.02**
Within 8 Hours			
Loss	577	0.80**	-0.12**
Draw	605	0.84**	-0.09**
Win	581	1.20**	0.00**

Asymmetric Effect: Losses decrease sentiment and posting, win increases posts but loss results impact on emotions.

Temporal Correlation: Clear relationship between match events and sentiment shifts

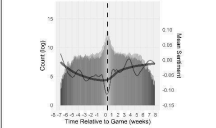
Real-Time Emotional Dynamics



Left: Aggregated sentiment post-match across all matches

Right: Individual match (Arsenal vs. Burnessmouth, 2021)

Below: Post count and sentiment relative to match (using



Key patterns:

- Universal sentiment drop at kick-off (toxicity)
- Diverging patterns based on final match outcome
- Negative reactions more sustained than positive
- Immediate measurable responses to key match events
- Sentiment is more acute to matches
- Sentiment decay over time following matches

Finding 3: Emotional Spillover Across Communities

Emotional states in football subreddits correlate to sentiment in unrelated subreddits

Time Period	Kendall's τ	n
All Match Posts	0.06***	275,863
During Matches	0.115***	234,624
Outside Matches	0.059***	241,839

Finding: Small correlation, but relative changes and statistical significance across large sample represent meaningful patterns of user behaviour at scale. The doubling of τ during matches suggests real-world events amplify cross-community emotional spillover. This represents (within this sample) thousands of additional, unique posts in unrelated subreddits.

	Negative	Neutral	Positive
Negative	80.0	-28.76	-28.73
Neutral	18.92	22.74	-7.59
Positive	-27.49	-5.12	11.66

Finding: Matching sentiment across communities occurs more than chance

Finding 4: Linguistic Spillover of Toxic Features

Method: Measured correlation between linguistic features in paired posts across subreddits.

Feature	Outside Match	During Match	Difference ($\Delta\tau$)
Profanity	0.061***	0.169***	0.108***
Violent words	0.022***	0.049***	0.027***
Interjections	0.059***	0.074***	0.015***
Exclamations	0.114***	0.154***	0.040***
All-together	0.052***	0.139***	0.087***

Finding: All linguistic toxicity markers strengthen during matches

Implication: Heightened emotional states from football events intensify toxic communication patterns across unrelated digital spaces

Implications & Applications

Digital communities are interconnected emotional ecosystems

For Research:

- Methods to measure emotional contagion, spreading
- Application beyond football (elections, breaking news)

For Platform Design:

- Predictive moderation during high risk events
- Early warnings through cross-platform monitoring
- Indicators for temporally linking cross-community posts

For Society:

- Understanding online-to-online harm pathways
- Hidden mechanisms of toxicity propagation

Broader Impact: Computational evidence of real-world event-driven emotional spillover across unrelated digital communities

ACL 2025 VIENNA JULY 27 - AUGUST 1

Contact: mark.j.hill@kcl.ac.uk
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Finding 3: Emotional Spillover Across Communities

Emotional states in football subreddits correlate to sentiment in unrelated subreddits

Time Period	Kendall's τ	n
All Paired Posts	0.085***	575,863
During Matches	0.118***	234,024
Outside Matches	0.059***	341,839

Finding: Small correlation, but relative changes and statistical significance across large sample represent meaningful patterns of user behaviour at scale. The doubling of τ during matches suggests real-world events amplify cross-community emotional spillover. This represents (within this sample) thousands of additional negative posts in unrelated subreddits.

	Negative	Neutral	Positive
Negative	40.00	-18.76	-26.73
Neutral	-18.92	22.74	-7.59
Positive	-27.49	-5.12	44.66

χ^2 standardised residuals

Finding: Matching sentiment across communities occurs more than chance

From Cultural Analytics to Big Data and Society

The promise of cultural computation

“At its core, this big data revolution is about how humans create and preserve a historical record of their activities. Its consequences will transform how we look at ourselves. It will enable the creation of new scopes that make it possible for our society to more effectively probe its own nature. Big data is going to change the humanities, transform the social sciences, and renegotiate the relationship between the world of commerce and the ivory tower.”

(Erez Aiden and Jean-Baptiste Michel, *Uncharted: Big Data as a Lens on Human Culture* 2013)

UNCHARTED

Big Data as a Lens
on Human Culture

Erez Aiden &
Jean-Baptiste Michel

The promise of data

“This is a world where massive amounts of data and applied mathematics replace every other tool that might be brought to bear. Out with every theory of human behavior, from linguistics to sociology. Forget taxonomy, ontology, and psychology. Who knows why people do what they do? The point is they do it, and we can track and measure it with unprecedented fidelity. With enough data, the numbers speak for themselves.”

(Chris Anderson, *Wired* 16.07)

 **WIRED**  [SUBSCRIBE](#)

CHRIS ANDERSON SCIENCE JUN 23, 2008 12:00 PM

The End of Theory: The Data Deluge Makes the Scientific Method Obsolete



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Analytics And Data Science

**Data Scientist: The Sexiest
Job of the 21st Century**

by Thomas H. Davenport and DJ Patil

From the Magazine (October 2012)



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Leaders | Regulating the internet giants

**The world's most valuable
resource is no longer oil, but
data**

The data economy demands a new approach to antitrust rules

Share



Traditional methods fail for datified/digital culture

“Rather than drawing on digital data to revitalize [social sciences’] practices, they [social scientists] are still trying to pass them off as new terrains for old methods. Cyberculture, virtual communities, online identities, computer-mediated communication: all these notions have been developed to **quarantine the novelty of electronic media**... It is no longer possible to circumscribe digital interactions to a specific sector, as their threads are now ubiquitous woven into the fabric of collective existence. Digital media offer much more than just another field to apply existing methods: they offer the possibility of **restructuring the study of social existence**.”

Tommaso Venturini and Bruno Latour, *The Social Fabric: Digital Traces and Quali-quantitative Methods* 2010

Cultural Analytics

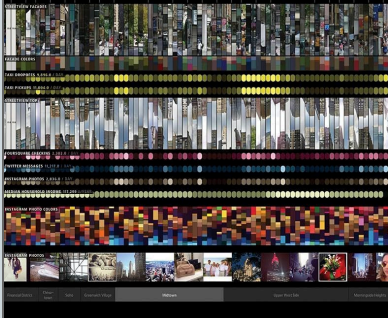
“Cultural analytics relies on the computer and search algorithms in order to discern and interpret culture.” This is done through “the exploration of large cultural data sets by means of interactive and intuitive visual analysis techniques” (Lev Manovich, *Cultural Analytics* 2009)

“[W]e are interested in combining both in the studies of cultures — focus on the particular, interpretation, and the past from the humanities and the focus on the general, formal models, and predicting the future from the sciences.”

(Lev Manovich, “The Science of Culture? Social Computing, Digital Humanities and Cultural Analytics” 2016)

LEV MANOVICH

CULTURAL ANALYTICS



So what can we actually analyse?

Meaning is contextually constituted

A cultural object does not carry its meaning inside itself.

- The same word means different things in different decades
- The same map means different things to a tourist and a resident
- The same book means different things on a syllabus and in a market

Hill, “Contextual Analytics: Reconstructing Meaning in Computational Cultural Analysis”, §2.1

What can computation operate on?

Computation needs a **discrete unit** — something countable, comparable, sortable.

Look back at the cat photograph and its metadata panel. The photograph is the cultural object. The panel — filename, date, camera, GPS — is what a computer can actually count and sort.

Turning a cultural object into such a unit is a **choice**, and the choice determines what you can subsequently see.

Five dimensions of context

Dimension	Unit of analysis	Native operations
Semiotic	the sign	frequency, co-occurrence, distributional similarity
Temporal	the timestamp	sequencing, periodisation, rate of change
Spatial	the coordinate	mapping, clustering, distance, centre—periphery
Social	the agent attribute	grouping, comparison across populations
Relational	the edge	centrality, paths, community detection

Hill, “Contextual Analytics”, §3 and Table 1

The claim is not the pattern

A method detects a **pattern**. A cultural claim needs a **second dimension** to interpret it.

Look again at the poster:

- **semiotic** — Reddit posts
- **social** — club communities
- **relational** — spillover between communities
- **temporal** — match times, 2008–2024

The finding is a *correlation of sentiment across communities*.

Note what the poster does **not** claim: that football causes toxicity.

Every week: *what did the method detect, and what am I adding when I interpret it?*

Which is also how you will write it up

Stage	Question
1. Object and claim	What am I studying, and what do I want to say about it?
2. Dataset and boundaries	What entered the data, what did not, and why?
3. Representing dimensions	What did I turn into a variable, and what did that discard?
4. Designing for interaction	Which dimensions does my claim actually depend on?
5. Interpretation and limits	What did I show, and what remains plausible but untested?

Hill, “Contextual Analytics”, §5 — and the structure of your 3,500-word commentary

Module details

Background Survey

Programming

- No experience 47%
- Basic experience 53%
- Languages used: R (35%), Python (24%), Other (59%)

Disciplines

- Media & Communications
- Digital Media
- Humanities (Literature, Philosophy, Art History)
- International Relations
- Business & Marketing

Statistics & Analysis

- Beginner level 94%
- Used summary statistics 59%
- Used hypothesis testing 41%
- Used visualisation 41%

Career Plans

- Industry 76%
- PhD 29%

- Public/Charity sector 24%
- Undecided 18%

Module plan

Lectures

- Each week the lecture introduces a key topic/concept from both a theoretical and practical perspective
- Following this introductory lecture, there will be live coding examples illustrating the topic in practice.
- There will be **weekly quizzes on KEATS, sat in class, which count for 40% of your final mark.**

Seminars

- Seminars are guided by weekly exercises that allow you to put the previous lectures topics into practice.
- You can do the exercises before the seminar, and answers will be released at the end of the week.

- Seminar time should be seen as a time to ask questions you may have regarding both lectures and exercises.

And weekly quizzes on KEATS!

 [Quiz: Week 1 - R data variables](#)

40% of your final mark

Sat in class, **nine times**,
each covering the previous
week.

Week 1's quiz is **practice
only** and does not count.

The quiz schedule

Week	Quiz covers	Week	Quiz covers
1	<i>practice — does not count</i>	7	Week 5
2	Week 1	8	Week 7
3	Week 2	9	Week 8
4	Week 3	10	Week 9
5	Week 4	11	Week 10
6	<i>reading week</i>		

Multiple choice, plus short pieces of code you write and run.

The other **60%** is a data analysis and commentary — 3,500 words including code ($\pm 15\%$) — due **21 January 2027**.

What is Python?

What is Python?

A General-Purpose Programming Language, Widely Used for Data Analysis

Key Features:

- Interactive interface language
— type commands, see immediate results
- Open-source and completely free — no licensing costs, ever
- Extensive community support
— thousands of contributors worldwide
- Constantly evolving — new analytical methods added regularly

Why Python for PGT Students?

- Incredibly useful for thesis and dissertation analysis
- Enables reproducible research (share and re-run

analyses)

- Highly valued in academia and industry
- Performs complex analyses impossible in spreadsheet software
- Unlike commercial software, accessible to all researchers globally

Positron: An Integrated Development Environment (IDE)

The Problem with Python on its own:

- Default Python interface is command-line only
- Difficult to navigate and manage files
- Limited visual feedback

The Positron Solution:

- Professional interface with organised panes
- Editor for writing and saving code, as scripts or notebooks
- Console for executing commands
- Variables pane showing your data and variables
- Integrated file browser, plot viewer, and help system

**Python is the engine,
Positron is the interface**

Positron is made by Posit, the same company that makes RStudio, so if you have seen RStudio the layout will look familiar.

Installation

Step 1: Install Python (3.12 or 3.13) from <https://www.python.org/downloads/>

- **Windows** — tick “**Add python.exe to PATH**” on the first screen. It is off by default
- **Mac** — the universal2 installer. Not the Python already on your Mac
- **Linux** — `python3.12` and `python3.12-venv` from your package manager

Step 2: Install Positron from <https://positron.posit.co/>

- **Windows** `.exe` · **Mac** `.dmg` (Apple silicon or Intel) · **Linux** `.deb` / `.rpm`

Step 3: Download `7AAVBCS5-setup.zip` from KEATS and unzip it somewhere sensible

Step 4: Create the environment in Positron

- **File** → **Open Folder** → choose the folder (the *whole* folder, not one file)
- **Ctrl+Shift+P** → **Python: Create Environment** → **Venv** → your Python
- Tick `requirements.txt` when offered, and wait

Only once, in Week 1 — the same environment covers all eleven weeks. Full instructions, including a two-command terminal version, are on KEATS.

Using Positron: The panes

Editor: write code here, in scripts or notebooks

Console: can type a single line of code here, but also where code executes, shows outputs and errors

Variables: overview of variables currently in memory, and their values

Files/plots/help: browse files, see visualisations you've created, output for documentation

Python as a Calculator

- Open Positron
- Find the Console pane
- Type: `2+3`
- Press Enter
- Result: `5`

```
1 2 + 3
```

```
5
```

What happened? You gave Python a command, Python executed it, Python returned the result.

Important: Python respects mathematical order of operations (PEMDAS/BODMAS)

Operation Symbols

Symbol	Operation	Example	Result
+	Addition	$5 + 3$	8
-	Subtraction	$10 - 4$	6
*	Multiplication	$7 * 8$	56
/	Division	$100 / 5$	20.0
**	Exponential	$2 ** 3$	8

^ is **not** exponentiation in Python. $2 ^ 3$ gives 6 — and gives no error.

Additional Useful

Operators:

% = Modulo (remainder):

$10 \% 3$ gives 1

// = Integer division: $10 // 3$ gives 3

Built-in Functions:

`abs()` = absolute value

`round()` = rounding

`math.sqrt()` = square root

`math.log()` = natural
logarithm

`math.pi` = built-in constant

π

Variables

- A labeled storage/memory location for information
- Allows you to save and reuse results
- Foundation of all programming

You assign variables with =

Naming rules and best practices:

- Must start with a letter (or an underscore)
- Can contain letters, numbers, underscores (`_`)
- Case-sensitive: `MyVar` `!=` `myvar` `!=` `MYVAR`
- No spaces
- Cannot use reserved words: `True`, `False`, `None`, `if`, `else`, `for`, `class`
- Use descriptive names: `student_age` not `x`
- Use `snake_case` (`my_variable_name`) — the Python convention

Variable Assignment Practice

```
1 temperature_c = 20
2 temperature_f = temperature_c * 9 / 5 + 32
3 temperature_f
```

68.0

- Build complex analyses step-by-step
- Reuse values without recalculating
- Make code readable and self-documenting
- Variables appear in the **Variables** pane

Data Types in Python: The Four Basic Types

1. **float** — For numbers with decimals

2. **int** — For whole numbers

- Uses less memory than float

N.B.: Rarely need to specify explicitly between int and float. Python decides for you, and converts when it needs to.

3. **str (String)** — For text

- Must use quotes (single or double)

4. **bool (Boolean)** — **True** or **False**

- Used for conditions and filtering
- Must be **capitalised**, no quotes
- Cannot be abbreviated to **T** / **F**

```
1 print(type(7), type(7.5), type("London"), type(True))
```

```
<class 'int'> <class 'float'> <class 'str'> <class 'bool'>
```

Data Types in Python: Checking and Converting Types

Data types can be converted using the following functions:

```
1 print(int("42"), type(int("42")))
2 print(float("3.14"), type(float("3.14")))
3 print(str(42), type(str(42)))
4 print(bool(0), bool(1))
```

```
42 <class 'int'>
3.14 <class 'float'>
42 <class 'str'>
False True
```

Check a type with `type()`:

```
1 answer = 42
2 type(answer)
```

```
int
```

Best Practices and Getting Help

Use Consistent Style:

- Pick naming convention and stick to it
- **Indentation is part of Python's syntax** — four spaces inside functions and loops, and it is not optional
- Add blank lines between logical sections
- Makes code easier to read and debug

Comment Your Code:

- Explain WHY, not just WHAT
- Good: `# Convert temperature for US audience`
- Less useful: `# Multiply by 9/5 and add 32`
- Future-you will thank present-you

Test as You Go:

- Run code chunk-by-chunk

- Don't write 100 lines then hope it works
- Fix errors immediately when they appear

Before Seeking Help:

- Check for typos (most common error!)
- Read the error message carefully
- Try Google with exact error message
- Create minimal example that reproduces problem

And ask me questions!

I'm certain you will ask me things I don't know the answer to. When that happens, I'll look up the answer. In coding — especially starting out — there are no stupid questions (honestly) — even professional coders Google constantly.

That's it! Thanks!

Questions?



Speaker notes